

Full-Stack AI Development — Internship Task Brief

Project: AI-Powered Professor & Scholarship Matching Platform

Issued by NeuroWebLabs | Track: Full-Stack + AI/LLM Engineering | Deadline: **September 30, 2026** | Support: engrnadeem26@gmail.com

1. Overview

Every year, thousands of students manually search for universities, shortlist professors, read their publications, and cold-email them for research positions, RAs, or PhD/MS sponsorship — a slow, repetitive process that could largely be automated. This task asks you to design and build a full-stack, AI-powered web application that does this work for the student: given a **field of interest (domain)** and a **target country**, the system finds top universities, identifies relevant professors, reads their recent research, scores how well each professor's work aligns with the student's interest, and produces a ready-to-use outreach list — including the option to send introduction emails automatically.

This is a real product opportunity, not just a training exercise. You are free to keep the finished application for your own use, sell it as a one-off tool, or turn it into a subscription-based SaaS product for students applying abroad.

2. Problem Statement

- Students spend weeks manually browsing university websites and faculty pages.
- It is hard to judge, at a glance, whether a professor's current research actually matches a student's interests.
- Finding correct, up-to-date faculty emails and contact details is tedious and error-prone.
- Most students write generic, low-response-rate cold emails with no idea which professors are actively funding/sponsoring students.
- There is no single place that combines university discovery + professor discovery + research-fit scoring + outreach in one flow.

3. Objective

Build a web application where a student enters their **domain/field of study** and **preferred country**. The system should then automatically:

- Identify the top universities in that country relevant to the domain.
- Discover professors in the relevant departments at those universities.
- Pull each professor's research focus, recent publications/papers, and lab/group info.
- Use an LLM to score and explain how well each professor's research aligns with the student's stated interest.
- Compile a clean, ranked CSV/list of matched professors with name, university, email, research area, alignment score, and a short justification.
- Optionally generate and send a personalized outreach email to each matched professor, on the student's behalf.

4. Core Features & User Flow

Step 1 — Input

Student submits: domain/field (e.g. "Reinforcement Learning", "Renewable Energy"), target country, and optionally a short bio/CV or interest statement, and study level (MS/PhD/RA).

Step 2 — University Discovery

System searches/ranks top universities in the chosen country for that domain (via web search, ranking APIs, or curated datasets).

Step 3 — Professor Discovery

For each shortlisted university, the system crawls or queries faculty directories to find professors in matching departments.

Step 4 — Research Enrichment

For each professor, pull recent publications (Google Scholar / Semantic Scholar / OpenAlex / DBLP APIs), lab pages, and research summaries.

Step 5 — AI Alignment Scoring

An LLM compares the student's interest statement against each professor's research summary/papers and produces a 0–100 alignment score with a short explanation of the fit.

Step 6 — Ranked Output

Results are shown in-app as a sortable/filterable table, and exportable as a CSV: Professor, University, Department, Email, Research Area, Recent Papers, Alignment Score, Justification.

Step 7 — Automated Outreach

Student reviews/edits an AI-drafted email template, then the system sends personalized emails to selected professors (with tracking of sent/opened/replied status).

5. Suggested Technical Architecture

Layer	Suggested Approach
Frontend	React / Next.js — input form, results dashboard with filters, CSV download, email review screen.
Backend / API	FastAPI or Node.js/Express — orchestrates search, scraping, enrichment, and scoring pipeline as async jobs.
University & Professor Discovery	Web search APIs (SerpAPI/Bing/Google CSE), university ranking datasets, faculty directory scraping (Playwright/BeautifulSoup).
Research Data	Semantic Scholar API, OpenAlex API, Google Scholar (via scraping or SerpAPI), DBLP for CS-related fields.
AI / Alignment Engine	LLM (Claude/GPT) with RAG over each professor's papers/abstracts to generate an alignment score + explanation; embeddings for quick similarity pre-filtering before the LLM pass.
Automation Layer	n8n (or similar) to orchestrate the multi-step pipeline (search → scrape → enrich → score → email), handle retries, and schedule periodic scholarship/funding checks.

Email Sending	SMTP / SendGrid / Gmail API for outreach, with per-professor personalization, send-rate limiting, and reply/bounce tracking.
Database	PostgreSQL/MongoDB for storing universities, professors, papers, scores, and outreach status per student account.
Auth & Accounts	Standard auth (email/OAuth) so this can later support multiple students/subscriptions if turned into a SaaS.

6. Bonus / Stretch Features

These are optional but strongly encouraged — they add real product value and differentiate the tool:

- **Scholarship Radar:** Scrape/aggregate free and upcoming scholarships relevant to the student's domain and country, with deadline alerts.
- **Funding/Sponsorship Signal:** Flag which professors have a track record of funding/sponsoring international students (e.g. grant history, lab size, past international students in their group).
- **Email Reply Detection:** Track opens/replies and notify the student, with suggested follow-up timing.
- **Multi-domain / Multi-country Search:** Let a student compare shortlists across 2–3 countries at once.
- **Resume/SOP Matching:** Let the student upload their CV so alignment scoring also factors in their own background, not just a typed interest statement.
- **Weekly Digest:** Auto-refresh professor/paper data and email the student a digest of new matching researchers or scholarships.

7. Deliverables & Suggested Milestones

Milestone	Deliverable	Target
M1 — Planning & Design	Architecture diagram, DB schema, chosen APIs/tools, wireframes for input & results screens.	Week 1
M2 — Discovery Pipeline	Working university + professor discovery for at least 2 domains and 2 countries.	Week 2
M3 — Research Enrichment & Scoring	Paper/abstract retrieval + LLM alignment scoring with explanations, tested end-to-end.	Week 3
M4 — Dashboard & CSV Export	Full frontend dashboard, filters/sorting, working CSV export.	Week 4
M5 — Outreach Automation	Email drafting + automated sending with tracking, n8n (or equivalent) pipeline wired end-to-end.	Week 5
M6 — Polish & Handover	Bug fixes, README/setup docs, demo video, and a short write-up of one bonus feature implemented.	By Sep 30, 2026

8. Final Submission Checklist

- Deployed link or fully runnable local build (with setup instructions).
- GitHub repository with clean commit history.
- Sample CSV output for at least one domain + country combination.
- Short demo video (5–7 minutes) walking through the flow end-to-end.
- Documentation of which APIs/tools were used and any limitations.

9. Evaluation Criteria

Criteria	Weight
Correctness & quality of university/professor discovery	20%
Quality & explainability of AI alignment scoring	25%
Full-stack execution (frontend UX, backend design, DB)	20%
Automation pipeline (n8n / job orchestration / reliability)	15%
Email outreach implementation & personalization	10%
Bonus features, polish, and documentation	10%

10. Business Potential

This is intentionally designed as more than a learning exercise. Interns are encouraged to treat it as a product they could own:

- **Personal use:** Use it yourself for grad school / scholarship applications.
- **One-off sales:** Sell the tool or the generated professor lists to students preparing to apply abroad.
- **Subscription SaaS:** Offer tiered plans (e.g. limited searches free, unlimited + automated outreach + scholarship alerts as a paid tier).
- **Service model:** Offer it as a "done-for-you" professor outreach service, charging per completed shortlist or per successful response.

11. Support & Questions

Tools, frameworks, and specific APIs are your choice — n8n, LangChain/LangGraph, or a fully custom pipeline are all acceptable as long as the end-to-end flow works reliably. If you get stuck on architecture, API selection, or scope, reach out directly:

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